

Tien Tran

tientran@buffalo.edu | (716)-295-3829 | linkedin.com/in/tmtien3004 | TMTien3004.github.io

Education

University at Buffalo

August 2022 – May 2026

B.A Computing and Applied Mathematics & B.S Computer Science

- **Overall GPA:** 3.8
- **Relevant Grad Coursework:** Intro Numeric Analysis I, Methods of Applied Maths I, Intro to Machine Learning.
- **Relevant Undergrad Coursework:** Intro to Numerical Analysis I, Introductory Linear Algebra, Survey of Partial Differential Equations, Intro to Differential Equations, Data-Oriented Computing for Mathematics, Scientific and Mathematical Computing, Intro to Probability.

Honors & Awards

Woepfel Research Initiative in Mathematics Scholarship

\$4,250

Dean's List

Fall 2022 – Present

Member of Phi Beta Kappa Honor Society

Spring 2025

Research Experience

Dept. of Mathematics, Undergraduate Researcher

June 2025 – Dec 2025

Advisor: Prof. Daozhi Han

- Programmed numerical algorithms to simulate and analyze the long-term behavior of nonlinear dynamical systems.
- Investigated the difference between high-order methods and low-order methods in capturing the chaotic system's statistical properties (mean, variance, statistical Cauchy difference).
- Designed a modular simulation framework to compute large-scale numerical solutions of the Lorenz 96 model using Runge-Kutta and forward Euler methods.

Dept. of Mathematics, Directed Reading Program

Sept 2025 – Dec 2025

Advisor: PhD Candidate Sayantan Sarkar

- Studied the mathematical theory of the Finite Element Method (FEM) for the numerical solution of partial differential equations.
- Derived weak formulations and implemented computational solvers using the FEniCSx/DOLFINx framework to model Poisson, heat, and hyperelastic equations.

Polymath Jr. Program, Undergraduate Researcher

June 2025 – Aug 2025

Advisor: Prof. Sean Howe

- Utilized SageMath to computationally evaluate the σ -moment generating functions for cyclic, dihedral, and generalized symmetric groups.
- Summarized an advanced mathematical research paper by creating a Wiki page on symmetric functions and commutative monoid rings.
- Collaborated with multiple undergraduate students to compute the Λ -distribution of various random matrix groups and presented the results at the Polymath Jr.'s conference.

Dept. of Computer Science & Engineering, Undergraduate Researcher

Aug 2024 – Dec 2024

Advisor: Prof. Roshan Ayyalasomayajula

- Built multiple data retrieval and manipulation tools using **PySpark** and **Pandas**.
- Synchronized filtered data between **MQTT** and **MinIO** database to ensure proper data storage and collection.
- Collaborated with other team members to debug the navigation mobile app, ensuring compatibility and optimal performance.

Dept. of Mathematics, Directed Reading Program

Jan 2024 – May 2024

Advisor: PhD Student Quoc Nguyen

- Investigated various types of data structures and path-finding algorithms to optimize the runtime of the Traveling Salesman Problem.
- Discussed new reading ideas with team members weekly and carried out a cumulative research presentation at a small conference in the Math Department.

Teaching Experience

Teaching Assistant, Department of Computer Science – Buffalo, NY

CSE 331 - Algorithms and Complexity

Jan 2025 – Present

- Conducted weekly office hours and recitations, providing support to students in understanding algorithms and mathematical proof techniques.
- Organized a review session for **20+** students, focusing on enhancing their mathematical proof-writing skills and problem-solving strategies.
- Evaluated and provided detailed feedback on assignments and exams for **200+** students.

CSE 241 - Digital Systems

Jan 2025 – May 2025

- Facilitated weekly lab sessions and open-hour support for **20+** students, guiding them in designing and implementing digital circuits.
- Maintained and managed lab equipment, ensuring a seamless learning environment for all students.

CSE 341 - Computer Organization

May 2024 – Dec 2024

- Prepared multiple practice problems and answered key documents for students by using LaTeX.
- Lead multiple weekly recitation sections of **30+** students by teaching new concepts in computer hardware and solving weekly homework problems.

Mathematics Grader, Department of Mathematics – Buffalo, NY

May 2024 – Dec 2024

- Cooperated with multiple professors to grade **200+** Calculus exam papers.
- Provided detailed feedback to students to aid in their understanding of course materials.

Academic Tutor, Tutoring and Academic Support Services – Buffalo, NY

Apr 2023 - May 2024

- Assisted **40+** students in solving homework problems and improving learning skills.
- Hosted one-on-one tutoring sessions for students who needed help with Intro to Computer Science, Calculus, and Linear Algebra assignments.

Projects

Self-Driving Car Simulation

Javascript, HTML, CSS

- Developed a car that can automatically drive through **20+** obstacles with **80%** accuracy.
- Implemented neural network techniques such as feedforward and genetic algorithms through JavaScript.
- Coded an HTML and CSS simulation of how the car can drive through other objects.

Neural Network Library

Python

- Constructed a neural network library from scratch based on "Neural Network from Scratch in Python" by Harrison Kinsley.
- Trained a machine learning model with **400+** steps and return an output of **90%** accuracy and predicted **9000+** images.
- Redesigned multiple feedforward and backpropagation algorithms, activation functions, and gradient descent optimization algorithms by using Python and Numpy.

Computer Vision Projects

Python

- Created three smaller computer vision projects utilizing Python and OpenCV library.
- Applied computer vision techniques such as hand gesture recognition with **90%** accuracy to control the computer's volume, draw on a canvas, and perform virtual mouse clicks.

Technical Skills

Languages: Python, Java, C/C++, Kotlin, Verilog, MIPS/ARM Assembly, JavaScript, HTML/CSS.

Technologies: Jupyter Notebook, VSC, Git, IntelliJ, Android Studio.

Libraries: NumPy, Pandas, Matplotlib, FEniCSx/DOLFINx, SKLearn, PySpark, SageMath, OpenCV.